



CVL2001: Climate Change and Adaptation

August 3, 2026

Lecture 1: Weather and Climate

Instructor for **Section B**: Prof. Joseph V. Puthussery

Department of Civil and Environmental Engineering

Class Schedule: Mondays and Thursdays

Class Timings: 12:00 noon – 1:00 PM LH 114

Email: josephvp@civil.iitd.ac.in

Course Content

- Difference between climate and weather
- Factors influencing regional and global climate
- Greenhouse gas effect
- Emissions of short-lived climate forcers (SLCPs)
- Climate relevant properties of climate forcers
- Radiative forcing
- Historical change in temperature and rainfall
- Working and goals of organizations related to climate change assessment (e.g. IPCC, UNEP etc.)
- Current mitigation policies
- Challenges in climate change adaptation

References

- Dessler, A. E. (2021). Introduction to Modern Climate Change, 3rd ed., Cambridge University Press.
- IPCC (2023). Climate Change 2023: AR6 Synthesis Report.

Course Evaluation Policy

Component	Weightage
Mid-semester examination	30%
Quizzes	30%
Major examination	35%
Attendance and class participation (min > 75% attendance)	5%
Total	100%

General Instructions

- For a re-minor or re-major examination, the student must inform the instructor within one week of the original examination.
- Examinations will be conducted using pen and paper.
- Minimum passing marks: 30% for both audit and credit students.
- Any violation of the Honor Code during an examination will invite disciplinary action.
- Any change in the course schedule or assessment plan will be communicated through the official course platform.



CVL2001: Climate Change and Adaptation

August 3, 2026

Lecture 1: Weather and Climate

The United States Left the Paris Agreement

27 January 2026

What changed?

- Climate science?
- Climate policy?
- Future emissions?
- All three?



“Can a country leave a climate agreement without changing climate science?”

Distinguishing Climate Science, Risk and Policy

1. CLIMATE SCIENCE: What is changing, and why?
2. CLIMATE RISK: Who or what may be affected?
3. CLIMATE POLICY: What actions should society take?



Scientific evidence informs policy decisions, but policy also involves economic, political and social choices.

Extreme Heat in Delhi: Same Hazard, Different Risks

Delhi/NCR Heatwave (June 2025) : Temperatures reached approximately 45°C in parts of the city.

Who faces the greatest risk?

- Office worker — reliable indoor cooling
- Delivery rider — prolonged exposure on roads
- Construction worker — heavy work in direct sunlight
- Older resident — health and cooling constraints

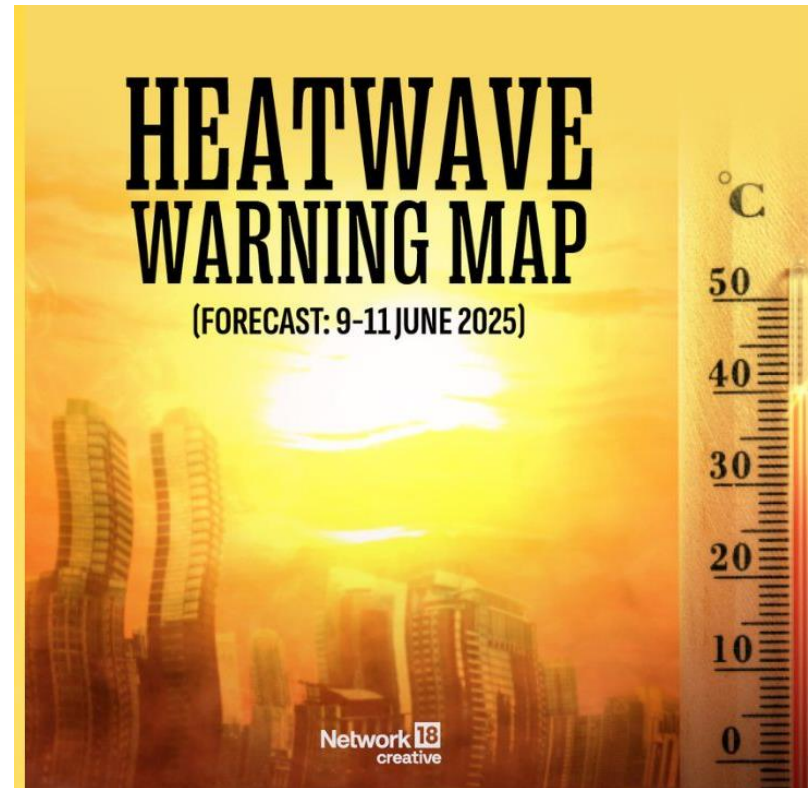


Image source: <https://www.cnbcvt18.com/india/environment/weather-update-to-day-imd-issues-heatwave-alert-for-delhi-ncr-rajasthan-punjab-haryana-sri-ganganagar-records-47-degrees-celsius-highest-temperature-19617507.htm>

Climate risk depends on:
Hazard + Exposure + Vulnerability

Weather and Climate: What Is the Difference?

WEATHER

- Atmospheric conditions at a particular time and place
- Changes over minutes, hours or days
- Example: Delhi's temperature today

CLIMATE

- Statistical description of weather over a long period
- Includes averages, variability and extremes
- Example: The frequency of heatwaves in Delhi over several decades

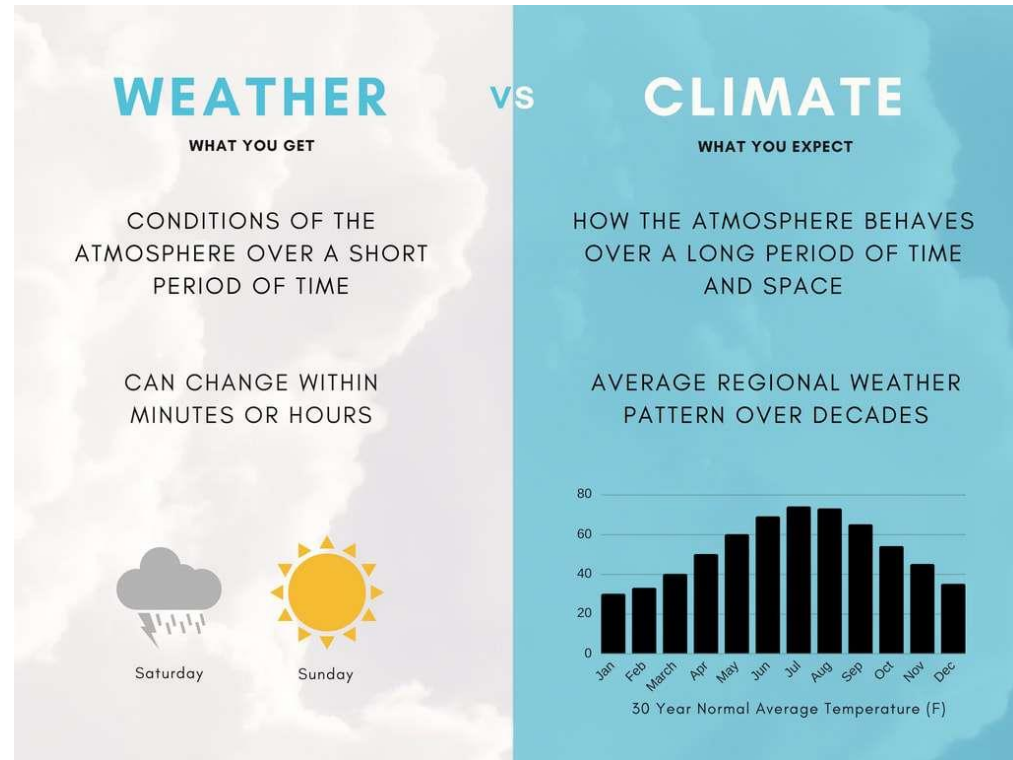


Image source: <https://norcast.tv/wp-content/uploads/2024/01/1672180073681.jpeg>

Weather is one event. Climate describes the pattern of possibilities.

What Is Climate Change?

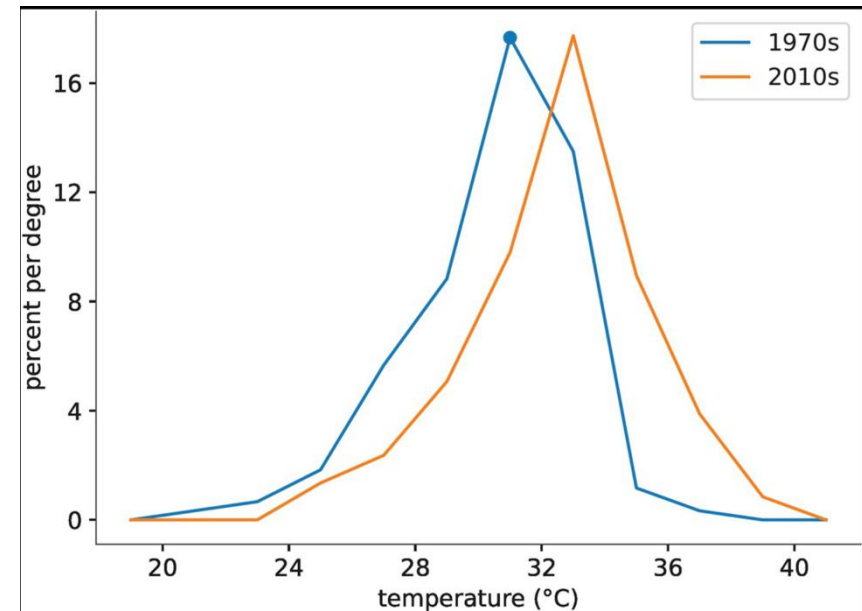
Climate change is: “**A systematic change in the long-term statistics of climate elements, sustained over several decades or longer.**”

Climate change may include:

- Changes in average temperature, rainfall, humidity or wind
- Changes in the variability of temperature and rainfall
- Changes in the frequency, intensity or duration of extreme events
- Changes in seasonal patterns, such as the timing of monsoon rainfall
- A single unusual weather event is not, by itself, climate change.

Houston: A Shift in the Temperature Distribution

- Daily maximum temperatures in September 1970s compared with the 2010s
- Most likely temperature increased from approximately 31°C to 33°C
- The distribution shifted by approximately 1.5°C
- Days above 34°C increased from 3% to 35%
- A small shift in the average produced a large increase in very hot days.



Source: Dessler (2021), Figure 1.1; data from Met Office HadISD.

How Do Scientists Identify the Causes of Climate Change?

- Detecting change
 - Long-term observations show whether the climate has changed.
- Attributing change
 - Physical mechanisms explain how different factors affect climate.
 - Climate models test the effects of natural and human influences.
 - Multiple independent lines of evidence are compared.

Detection asks: Has the climate changed?

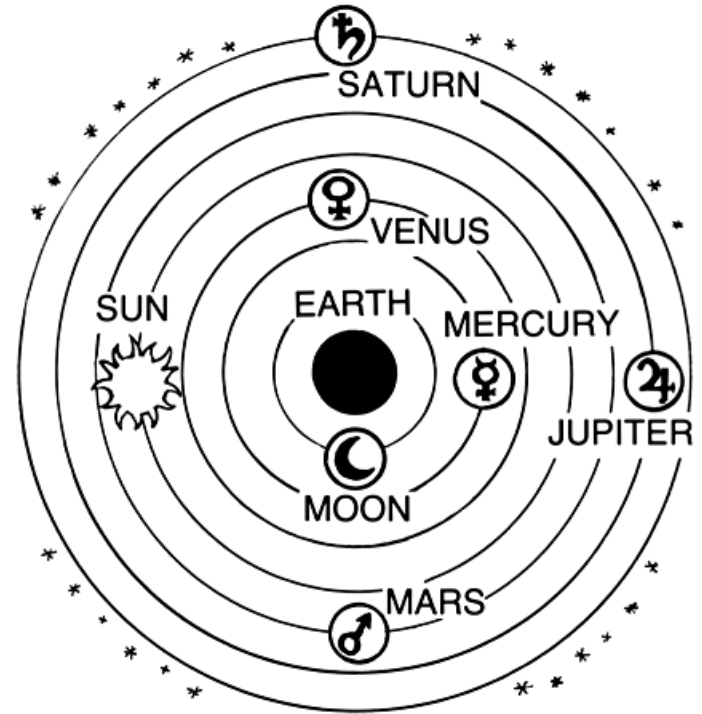
Attribution asks: What caused the change?

Can a Model Be Useful but Still Be Wrong?

The Ptolemaic model placed Earth at the centre of the planetary system.

- It successfully predicted many planetary positions.
- Some predictions were remarkably accurate.
- However, its explanation of the planetary system was incorrect.

A correct prediction does not prove that every part of a model is correct.



https://study.com/cimages/multimages/16/ptolemaic_system_psf7999927724135214065.png

How should scientists decide whether to accept, revise or reject a model?

How Scientific Explanations Become Reliable

Scientific explanations are strengthened through:

- Testing against observations
- Peer review and critical evaluation
- Replication by independent researchers
- Comparison with alternative explanations
- Consistency across multiple lines of evidence

Scientific confidence comes from repeated scrutiny—not from the authority of one individual.

What Is Causing the Climate to Change?

1. NATURAL FACTORS

- Changes in solar energy
- Volcanic eruptions
- Natural climate variability

2. HUMAN ACTIVITIES

- Burning fossil fuels
- Deforestation and land-use change
- Agriculture and industrial activities

Current global warming is primarily caused by human emissions of greenhouse gases.

Key Takeaways

- Weather describes short-term atmospheric conditions.
- Climate describes long-term weather statistics.
- Climate change is a systematic change sustained over several decades or longer.
- A small change in average conditions can produce a large change in extremes.
- Climate may change because of natural processes and human activities.
- Scientific explanations become reliable through repeated testing and scrutiny.